

Unified VDS/DVP/BH Number System Derivation with Ramanujan Acceleration

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2026-04-16

PAPER_1108: Unified VDS/DVP/BH Number System Derivation with Ramanujan Acceleration

Abstract

We present a unified derivation engine for the three UQFF number systems — Vacuum Density Series (VDS), Dipole Vortex Primes (DVP), and Buoyancy Harmonics (BH) — with Ramanujan acceleration and cross-validation. These three number systems collectively govern the vacuum structure, vortex quantization, and charge-reactivity gravity (U_{g2}) of the UQFF framework.

1. VDS: Vacuum Density Series

Definition

$$\text{VDS} = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} = \text{Li}_{26}([\text{SSq}])$$

where Li_{26} is the polylogarithm of order 26.

Ramanujan Acceleration

The convergence is accelerated by the factor $R_n^{(26,3)}$:

$$S_{26}^{(3)}([\text{SSq}]) = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} \cdot R_n^{(26,3)}$$
$$R_n^{(26,3)} = \frac{(2\pi)^{n/6}}{n!} \left[1 + \sum_{m=1}^3 \frac{1}{n^{26m}} \sum_{j=1}^{26} (-1)^{j+1} \binom{26}{j} \frac{(26-j)!}{n^j} \right]$$

This achieves 60+ digit precision in ≤ 50 terms.

Physical Meaning

VDS encodes the **vacuum density normalization** — the suppression of the cosmological vacuum energy through the 26-layer compactification structure. $[\text{SSq}] = 0.57$ ensures convergence and bounds the vacuum from above.

2. DVP: Dipole Vortex Primes

Definition

For primes $p > 26$:

$$a(p) = \frac{[\text{SSq}]^{\pi(p)}}{p^{26}}$$

where $\pi(p)$ is the prime-counting function.

Key Values

p	$\pi(p)$	$a(p)$
29	10	$\approx 3.62 \times 10^{-42}$
31	11	$\approx 4.92 \times 10^{-43}$
37	12	$\approx 1.80 \times 10^{-48}$
41	13	$\approx 2.60 \times 10^{-50}$
113	30	$\approx 10^{-83}$

Physical Meaning

DVP encodes **vortex quantization** — the proplyd orbital radius $r_q \approx 0.0973$ AU and Navier-Stokes bounded vorticity. The prime-indexed structure ensures each vortex mode corresponds to a unique topological quantum number.

3. BH: Buoyancy Harmonics

Definition

$$U_{g2} = \sum_{m=1}^{\infty} H_m \cdot (1 - e^{-[\text{SSq}] \cdot m}) \cdot \cos(\omega_{U_{g2}} \cdot t_n)$$

where $H_m = \sum_{k=1}^m 1/k$ is the m -th harmonic number.

Saturation Behavior

As $m \rightarrow \infty$: $(1 - e^{-[\text{SSq}] \cdot m}) \rightarrow 1$

The saturation factor monotonically approaches unity, ensuring U_{g2} is bounded by the harmonic series $\sum H_m \cos(\omega t)$.

Physical Meaning

BH drives **charge-reactivity gravity** (U_{g2}) and saturation harmonics in $E_{\text{net}}(t)$. The harmonic sum H_m introduces logarithmic growth modulated by [SSq]-suppressed saturation and temporal oscillation.

4. Cross-Validation

Internal Consistency

1. **VDS** $\leftrightarrow$ **DVP**: DVP samples the VDS polylogarithm at prime arguments. The sum $\sum_p a(p)$ is bounded by $\text{Li}_{26}([\text{SSq}])$.
2. **DVP** $\leftrightarrow$ **BH**: Vortex modes (DVP) seed the harmonic structure of U_{g2} (BH) through their topological quantum numbers.
3. **BH** $\leftrightarrow$ **VDS**: The vacuum density (VDS) normalizes the saturation parameter in BH, ensuring $[\text{SSq}]$ is self-consistent across all three systems.

Ratio Test

$$\frac{\sum_p a(p)}{\text{VDS}} \ll 1$$

confirms that prime-indexed modes are a sparse subset of the full vacuum series.

5. Implementation

Calculator: `VDS DVP BH Unified Number System Calculator` in `CondensedPhysics.py`

- Computes VDS with and without Ramanujan acceleration (configurable N terms)
- Evaluates DVP for the first 15 primes > 26
- Computes BH up to harmonic order M with configurable $\omega_{U_{g2}}$ and t_n
- Cross-validates DVP/VDS ratio and BH saturation limit

References

- Ramanujan, S. (1918). On certain trigonometrical sums. *Trans. Camb. Phil. Soc.* 22, 259.
- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.
- PAPER_491: MUGE Compressed Nine-Term Gravity Framework (VDS/DVP/BH canonical definitions).